21-242 Recitation 12

RREF — Q3. (T A)= (T (A))

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that $\ker(T_A) = \ker(T_{\mathrm{RREF}(A)})$.

Course-level solution

By Q2, $\mathrm{RREF}(A) = GA$ with G invertible. Then $A\vec{x} = \vec{0}$ iff $GA\vec{x} = \vec{0}$ (multiply by G or by G^{-1}). Thus $\ker T_A = \ker T_{GA} = \ker T_{\mathrm{RREF}(A)}$.